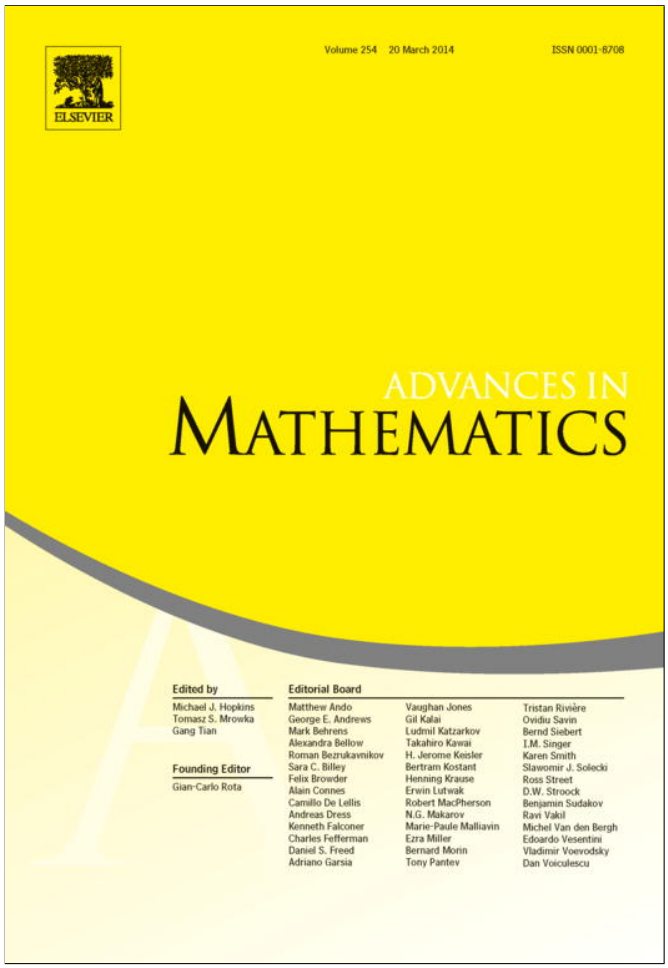
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The unramified Brauer group of norm one tori

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ABSTRACT

Let k be a number field and K/k Galois. We transform the construction of the unramified Brauer group of the norm one torus $\mathrm{R}^1_{K/k}(\mathbb{G}_m)$ into the construction of a special abelian extension over K. If $k=\mathbb{Q}$ and $K/\mathbb{Q}$ biquadratic, we explicitly construct the unramified Brauer group of $\mathrm{R}^1_{K/\mathbb{Q}}(\mathbb{G}_m)$.

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0. Introduction

Let T be a torus over a field k of characteristic zero, X a principal homogeneous space of T, and X^c a smooth compactification of X. Since the Brauer group $\mathrm{Br}(X^c) := H^2_{\acute{e}t}(X^c, \mathbb{G}_m)$ is a birational invariant of smooth proper varieties, it does not depend on the choice of X^c but only depends on X; it is called the unramified Brauer group of X. Let $\mathrm{Br}_0(X^c)$ be the image of the natural map $\mathrm{Br}(k) \to \mathrm{Br}(X^c)$. Formulas for

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$\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ can be found in [3]. In particular, if K/k is a Galois extension and $T = \mathrm{R}^1_{K/k}(\mathbb{G}_m)$ its norm one torus, then $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) \cong H^3(\mathrm{Gal}(K/k), \mathbb{Z})$. If K/k is cyclic or of prime degree (not necessarily Galois), then $\mathrm{Br}(X^c) = \mathrm{Br}_0(X^c)$.

It is well known that the Brauer–Manin obstruction to the Hasse principle and weak approximation for rational points is the only one on X^c [12]. To compute the Brauer–Manin obstruction, one needs to construct the Brauer group. Recently, Colliot-Thélène [2] gave an explicit construction for a multi-norm torus of dimension 5. However, for general tori, it is still open, even for the norm one torus $R^1_{K/\mathbb{Q}}(\mathbb{G}_m)$, where $K/\mathbb{Q}$ biquadratic.

The main aim of this article is to construct the unramified Brauer group for the torus $\mathrm{R}^1_{K/k}(\mathbb{G}_m)$ when K/k is Galois and $H^3(k, \mathbb{Z}) = 0$, e.g., $H^3(k, \mathbb{Z}) = 0$ holds for any number field k. In Section 1, we show any element in the unramified Brauer group of $\mathrm{R}^1_{K/k}(\mathbb{G}_m)$ has a form from cup-product. Furthermore, we transform the construction of the unramified Brauer group into the construction of a special abelian extension over K (see Theorem 5). Some applications and examples are also given in this section. In Section 2, using results of double coverings of $\mathbb{Q}^{ab}/\mathbb{Q}$ in [1,6,15], we give the explicit construction of the unramified Brauer group for the torus $R^1_{K/\mathbb{Q}}(\mathbb{G}_m)$, where $K/\mathbb{Q}$ biquadratic.

1. The Brauer group of $\mathrm{R}^1_{K/k}(\mathbb{G}_m)$ when K/k is Galois

Let k be a field with $char(k) = 0$. Let K/k be a finite extension of degree m and $T = \mathrm{R}^1_{K/k}(\mathbb{G}_m)$ its norm one torus. Suppose $K = k(\omega)$, then $\{1, \omega, \ldots, \omega^{m-1}\}$ is a basis of K as a k-vector space. Let $\Xi = x_0 + x_1\omega + \cdots + x_{m-1}\omega^{m-1}$, where $x_0, \ldots, x_{m-1}$ are variables. Let $X \subset \mathbf{A}^m_k$ be the affine k-variety defined by $N_{K/k}(\Xi) = \alpha \in k^\times$ (α is fixed), which is a principal homogeneous space of T. Then $\bar{k}[X]^\times/\bar{k}^\times \cong \widehat{T}$ as $\mathrm{Gal}(\bar{k}/k)$-modules, where $\bar{k}[X]^\times$ is the group of invertible regular functions of $\overline{X}$ and $\widehat{T}$ is the character group of T.

Let X^c be a smooth compactification of X. Let $\mathrm{Br}(X)$ (resp. $\mathrm{Br}(X^c)$) be the Brauer group of X (resp. X^c) and $\mathrm{Br}_0(X)$ (resp. $\mathrm{Br}_0(X^c)$) the image of $\mathrm{Br}(k)$ in $\mathrm{Br}(X)$ (resp. $\mathrm{Br}_0(X^c)$).

Lemma 1. *$\mathrm{Br}(X^c)$ is contained in the image of $H^2(k, \bar{k}[X]^\times)$ in $\mathrm{Br}(X)$.*

Proof. By the Hochschild–Serre Spectral sequence, we have

$$\mathrm{Pic}(\overline{X})^{\mathrm{Gal}(\bar{k}/k)} \to H^2\big(k, \bar{k}[X]^\times\big) \to \mathrm{Ker}\big[\mathrm{Br}(X) \to \mathrm{Br}(\overline{X})\big] \to H^1\big(k, \mathrm{Pic}(\overline{X})\big).$$

Since $\overline{X} \cong \mathbb{G}^d_m$ over $\bar{k}$, we have $\mathrm{Pic}(\overline{X}) = 0$, where $d = [K : k] - 1$. Therefore

$$H^2\big(k, \bar{k}[X]^\times\big) \cong \mathrm{Ker}\big[\mathrm{Br}(X) \to \mathrm{Br}(\overline{X})\big]. \tag{1}$$

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Since X^c is geometrically rational, we have $\mathrm{Br}(\overline{X^c}) = 0$. Therefore

$$\mathrm{Br}\big(X^c\big) \subset \mathrm{Ker}\big[\mathrm{Br}(X) \to \mathrm{Br}(\overline{X})\big]. \qquad \square$$

Let $\Gamma_k = \mathrm{Gal}(\bar{k}/k)$ and $\Gamma_K = \mathrm{Gal}(\bar{k}/K)$. Let $\mathbb{Z}[K/k]$ denote the group ring $\mathbb{Z}[\Gamma_k/\Gamma_K]$. For $i \geqslant 0$, we have the cup product

$$(\cdot\,,\cdot) : \mathbb{Z}[K/k] \times H^i(K,\mathbb{Z}) \to H^i\big(K,\mathbb{Z}[K/k]\big).$$

Let $\mathrm{Cor}_{K/k}$ be the corestriction map $H^i(K,\cdot\,) \to H^i(k,\cdot\,)$.

Lemma 2. *Let $\Gamma_K \in \Gamma_k/\Gamma_K$. Then $\mathrm{Cor}_{K/k}(\Gamma_K,\cdot\,) : H^i(K,\mathbb{Z}) \to H^i(k,\mathbb{Z}[K/k])$ is the inverse map of Shapiro's isomorphism $sh : H^i(k,\mathbb{Z}[K/k]) \to H^i(K,\mathbb{Z})$.*

Proof. The case $i = 0$ is obvious. Let $i > 0$. Let $g = \mathrm{Cor}_{K/k}(\Gamma_K,\cdot\,)$. Since sh is an isomorphism (see [10, Proposition I.1.6.3]), it is enough to show $sh \circ g = id$. Let $C^\bullet(\Gamma_K,\mathbb{Z})$ (resp. $C^\bullet(\Gamma_K,\mathbb{Z}[K/k])$) be the homogeneous cochain complex of Γ_K with coefficients in $\mathbb{Z}$ (resp. $\mathbb{Z}[K/k]$) (see [10, Chapter 1]). Suppose $x \in H^i(K,\mathbb{Z})$, choose an i-cocycle $u \in C^i(\Gamma_K,\mathbb{Z})$ which represents x.

We use notations in p. 46 of [10, Chapter I, Section 5], let $c = \Gamma_K\sigma \in \Gamma_K \setminus \Gamma_k$, we choose a fix representative $\bar{c} \in c$ and choose $\bar{c}$ to be the identity if $c = \Gamma_K \in \Gamma_K \setminus \Gamma_k$. For any $(\sigma_0,\ldots,\sigma_i) \in \bigoplus_{i+1} \Gamma_K$, then

$$\begin{aligned} g(u)(\sigma_0,\ldots,\sigma_i) &= \mathrm{Cor}_{K/k}(\Gamma_K,u)(\sigma_0,\ldots,\sigma_i) \\ &= \sum_{c\in\Gamma_K\setminus\Gamma_k} u\big(\bar{c}\sigma_0\overline{c\sigma}_0^{-1},\ldots,\bar{c}\sigma_i\overline{c\sigma}_i^{-1}\big)\bar{c}^{-1}\Gamma_K. \end{aligned}$$

Let j_{Γ_K} be the projection $\mathbb{Z}[K/k] \to \mathbb{Z}$ by $\Gamma_K \mapsto 1, \gamma\Gamma_K \mapsto 0$ if $\gamma \notin \Gamma_K$. Since $\sigma_i \in \Gamma_K$ for any i, we have $\Gamma_K = \Gamma_K\sigma_i \in \Gamma_K \setminus \Gamma_k$, hence $\overline{c\sigma}_i$ is the identity for $c = \Gamma_K \in \Gamma_K \setminus \Gamma_k$. Therefore

$$\begin{aligned} \big(sh \circ g(u)\big)(\sigma_0,\ldots,\sigma_i) &= j_{\Gamma_K}\Bigg(\sum_{c\in\Gamma_K\setminus\Gamma_k} u\big(\bar{c}\sigma_0\overline{c\sigma}_0^{-1},\ldots,\bar{c}\sigma_i\overline{c\sigma}_i^{-1}\big)\bar{c}^{-1}\Gamma_K\Bigg) \\ &= u(\sigma_0,\ldots,\sigma_i). \qquad \square \end{aligned}$$

Let Ξ be as in the beginning of this section, we can see $\Xi \in \bar{k}[X]^\times$. For any character χ of Γ_K, it is a homomorphism from Γ_K to $\mathbb{Q}/\mathbb{Z}$, which gives rise to an element of $H^1(K,\mathbb{Q}/\mathbb{Z})$ and then $H^2(K,\mathbb{Z})$. Then the cup-product (Ξ,χ) gives an element of $H^2(K,\bar{k}[X]^\times)$. Let $\mathrm{Cor}_{K/k} : H^2(K,\bar{k}[X]^\times) \to H^2(k,\bar{k}[X]^\times)$ be the corestriction. Then (Ξ,χ) gives the element $\mathrm{Cor}_{K/k}(\Xi,\chi) \in H^2(k,\bar{k}[X]^\times) \subset \mathrm{Br}(X)$. The following Proposition will show any element of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ has this form.

Proposition 3. *Let k be a field of characteristic 0 and let χ be a character of Γ_K.*

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(a) *The element* $\mathrm{Cor}_{K/k}(\Xi,\chi)=0\in \mathrm{Br}(X)/\mathrm{Br}_0(X)$ *if and only if* χ *is the restriction of a character of* Γ_k.
(b) *Suppose* $H^3(k,\mathbb{Z})=0$, *e.g.,* k *is a number field or p-adic number field (see* [9, Corollary 4.7]). *Each element of* $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ *in the image of* $H^2(k,\bar{k}[X]^\times)$ *is of the form* $\mathrm{Cor}_{K/k}(\Xi,\chi)$.

Proof. Using the natural exact sequence

$$0\to\mathbb{Z}\to\mathbb{Z}[K/k]\to\widehat{T}\to 0,$$

we deduce the following exact sequence

$$H^2(k,\mathbb{Z})\to H^2\big(k,\mathbb{Z}[K/k]\big)\xrightarrow{f}H^2(k,\widehat{T})\to H^3(k,\mathbb{Z}). \tag{2}$$

Define the Γ_k-morphism $j:\mathbb{Z}[K/k]\to\bar{k}[X]^\times$ by $\Gamma_K\mapsto\Xi$. Then we have the maps $\mathbb{Z}[K/k]\to\bar{k}[X]^\times\to\bar{k}[X]^\times/\bar{k}^\times\ (\cong\widehat{T})$. Therefore we have the induced map $H^2(k,\mathbb{Z}[K/k])\to H^2(k,k[X]^\times)\xrightarrow{g}H^2(k,\widehat{T})$, and the composite map $H^2(k,\mathbb{Z}[K/k])\to H^2(k,\widehat{T})$ is coincide with the map f in (2).

By the basic exact sequence $0\to\bar{k}\to\bar{k}[X]^\times\to\bar{k}[X]^\times/\bar{k}^\times(\cong\widehat{T})\to 0$, we have the following exact sequence

$$\mathrm{Br}(k)\to H^2\big(k,\bar{k}[X]^\times\big)\xrightarrow{g}H^2(k,\widehat{T}). \tag{3}$$

Hence g induces an injection $g':H^2(k,\bar{k}[X]^\times)/\mathrm{Ima}[\mathrm{Br}(k)]\hookrightarrow H^2(k,\widehat{T})$, where $\mathrm{Ima}[\mathrm{Br}(k)]$ is the image of $\mathrm{Br}(k)$ in $H^2(k,\bar{k}[X]^\times)$. We have the following commutative diagram

$$\begin{array}{ccc} H^2\big(k,\mathbb{Z}[K/k]\big) & \to & H^2\big(k,\bar{k}[X]^\times\big)/\mathrm{Ima}\big(\mathrm{Br}(k)\big) \\ & \searrow^{f} & \downarrow g' \\ & & H^2(k,\widehat{T}). \end{array} \tag{4}$$

Since g' is injective, by diagram (4), the exact sequence (2) gives the following exact sequence

$$H^2(k,\mathbb{Z})\to H^2\big(k,\mathbb{Z}[K/k]\big)\to H^2\big(k,\bar{k}[X]^\times\big)/\mathrm{Ima}\big(\mathrm{Br}(k)\big). \tag{5}$$

By Lemma 2 and functoriality of cup product, we have the following commutative diagram

$$\begin{array}{ccccc} & & H^2(K,\mathbb{Z}) & & \\ & \overset{\mathrm{Res}_{k/K}}{\nearrow} & \downarrow g_1 & \searrow^{g_2} & \\ H^2(k,\mathbb{Z}) & \to & H^2\big(k,\mathbb{Z}[K/k]\big) & \to & H^2\big(k,\bar{k}[X]^\times\big), \end{array}$$

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where $g_1 = \mathrm{Cor}_{K/k}(\Gamma_K, \cdot)$ is the inverse of Shapiro's isomorphism (by Lemma 2), $g_2 = \mathrm{Cor}_{K/k}(\Xi, \cdot)$ and the lower right map is induced by j with $j(\Gamma_K) = \Xi$. By this commutative diagram, (5) is identity with the following exact sequence

$$H^2(k, \mathbb{Z}) \xrightarrow{\mathrm{Res}_{k/K}} H^2(K, \mathbb{Z}) \xrightarrow{h} H^2\big(k, \bar{k}[X]^\times\big)/\operatorname{Ima}\big[\mathrm{Br}(k)\big], \tag{6}$$

where $h = \rho \circ \mathrm{Cor}_{K/k}(\Xi, \cdot)$ and ρ is the quotient map

$$H^2\big(k, \bar{k}[X]^\times\big) \to H^2\big(k, \bar{k}[X]^\times\big)/\operatorname{Ima}\big[\mathrm{Br}(k)\big].$$

Since the map $H^2(k, \bar{k}[X]^\times) \to \mathrm{Br}(X)$ is injective by (1), the map

$$H^2\big(k, \bar{k}[X]^\times\big)/\operatorname{Ima}\big[\mathrm{Br}(k)\big] \to \mathrm{Br}(X)/\mathrm{Br}_0(X)$$

is also injective. By the exact sequence (6), we immediately have $\mathrm{Cor}_{K/k}(\Xi, \chi)$ is zero in $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ if and only if χ is the restriction of a character of Γ_k. Then we prove part (a).

Now we prove part (b). Since $H^3(k, \mathbb{Z}) = 0$, the map f in sequence (2) is surjective. The following diagram is commutative

$$\begin{array}{ccc} H^2\big(k, \mathbb{Z}[K/k]\big) & \longrightarrow & H^2\big(k, \bar{k}[X]^\times\big) \\ & \searrow^{f} & \downarrow g \\ & & H^2(k, \widehat{T}). \end{array}$$

Then g is also surjective since f is surjective. The sequence (3) induces f': $H^2(k, \bar{k}[X]^\times)/\operatorname{Ima}[\mathrm{Br}(k)] \to H^2(k, \widehat{T})$ is an isomorphism. By the sequence (2) and diagram (4), the second map in (5) is also surjective, hence the map h in (6) is also surjective. Therefore each element of $H^2(k, \bar{k}[X]^\times)/\operatorname{Ima}[\mathrm{Br}(k)]$ is of the form $\mathrm{Cor}_{K/k}(\Xi, \chi)$ by (6), where χ is a character of Γ_K. Therefore each element of $\mathrm{Br}(X)/\mathrm{Br}_0(X)$ in the image of $H^2(k, \bar{k}[X]^\times)$ is of the form $\mathrm{Cor}_{K/k}(\Xi, \chi)$. □

In the following we will give the description of the unramified Brauer group of X. First we give some notation.

Suppose K/k is Galois. *We say a field extension L/K satisfies the condition* $(*)$ *over k if*:

L/K' is abelian for any subfield $K' \subset K$ containing k and satisfying K/K' is cyclic.

Obviously L/K is abelian if L/K satisfies the condition $(*)$ over k.

Lemma 4. *Suppose K/k is Galois and L/K satisfies the condition $(*)$ over k. Then L/k is Galois and $\mathrm{Gal}(L/K)$ is contained in the center of $\mathrm{Gal}(L/k)$.*

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Proof. We will show $\sigma(L) = L$ for any $\sigma \in \Gamma_k$. Let K^σ be the fixed subfield of σ in K. Since K/K^σ is cyclic, we have L/K^σ is abelian by the condition $(*)$ over k. Since $\sigma \in \Gamma_{K^\sigma}$, it implies $\sigma(L) = L$. Therefore L/k is Galois.

Let $\sigma \in \mathrm{Gal}(L/K)$. For any $g \in \mathrm{Gal}(L/k)$, let K^g be the subfield of K fixed by g. Hence K/K^g is cyclic. By the condition $(*)$ over k, $\mathrm{Gal}(L/K^g)$ is abelian. This implies $\sigma g = g\sigma$ since $\sigma, g \in \mathrm{Gal}(L/K^g)$, hence σ is contained in the center of $\mathrm{Gal}(L/k)$. □

Theorem 5. *Let k be a field of characteristic 0. Suppose K/k is Galois. Let χ be a character of Γ_K. Then:*

(a) $\mathrm{Cor}_{K/k}(\Xi, \chi) \in \mathrm{Br}(X^c)$ *if and only if χ can factor through an abelian extension L/K which satisfies the condition $(*)$ over k.*
(b) *If $H^3(k, \mathbb{Z}) = 0$, then any element of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ is of the form $\mathrm{Cor}_{K/k}(\Xi, \chi)$.*

Proof. Part (b) follows from Lemma 1 and Proposition 3. So it remains to prove part (a).

Let $k(X)$ be the function field of X. Let A be a discrete valuation ring containing k with fraction field $k(X)$ and residue field κ_A. There is a residue map

$$\partial_A : \mathrm{Br}\big(k(X)\big) \to H^1(\kappa_A, \mathbb{Q}/\mathbb{Z}).$$

By Grothendieck's purity theorem, we have

$$\mathrm{Br}\big(X^c\big) = \bigcap_A \mathrm{Ker}(\partial_A) \subset \mathrm{Br}\big(k(X)\big),$$

where A runs through all discrete valuation rings as above.

Suppose L/K satisfies the condition $(*)$ over k. Let χ be a character of Γ_K which factors through $\mathrm{Gal}(L/K)$. Let $B = \mathrm{Cor}_{K/k}(\Xi, \chi) \in \mathrm{Br}(X)$. We want to prove $B \in \mathrm{Br}(X^c)$. Therefore we only need to prove the triviality of $\partial_A(B)$ for any such discrete valuation ring A, *i.e.*, $\partial_A(B)(g) = 0$ for any $g \in \mathrm{Gal}(\bar{\kappa}_A/\kappa_A)$.

We extend the embedding $k \subset \kappa_A$ to an embedding $\bar{k} \subset \bar{\kappa}_A$, so that g acts also on $\bar{k}$. Let K^g be the subfield of K fixed by g, with cyclic Galois group $\mathrm{Gal}(K/K^g)$.

Since X is geometrically integral, the function field $K(X) \cong k(X) \otimes_k K$ of $X \times_k K$ is finite over $k(X)$. Since $k \subset A$ and K/k is finite (and étale), $A \otimes_k K$ is finite étale over A. We can extend A to a discrete valuation ring A_K of $K(X) = \mathrm{Frac}(A \otimes_k K)$ with residue field $\kappa_{A_K} = \kappa_A.K$. Indeed, the completion of $k(X)$ for the given valuation is isomorphic to $\kappa_A((\pi))$, where π is a uniformizer. Considering the valuation given by π on $(K.\kappa_A)((\pi))$ and using $K(X) = k(X) \otimes_k K$, we see that the completion $\widehat{K(X)}$ of $K(X)$ with respect to A_K is $\kappa_{A_K}((\pi))$. This defines a discrete valuation on $K(X)$ with valuation ring A_K.

For any intermediate field M of K/k, we have similarly the valuation ring A_M of $M(X)$ with residue field κ_{A_M}. We write A_g for A_{K^g}.

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By [5, Proposition 1.1.1], we have the commutative diagram

$$\begin{array}{ccc} \mathrm{Br}\big(k(X)\big) & \xrightarrow{\partial_A} & H^1(\kappa_A, \mathbb{Q}/\mathbb{Z}) \\ {\scriptstyle \mathrm{Res}_{k/K^g}}\downarrow & & \downarrow{\scriptstyle \mathrm{Res}_{\kappa_A/\kappa_{A_g}}} \\ \mathrm{Br}\big(K^g(X)\big) & \xrightarrow{\partial_{A_g}} & H^1(\kappa_{A_g}, \mathbb{Q}/\mathbb{Z}). \end{array}$$

Since $\kappa_{A_g} = \kappa_A.K^g$, we have $g \in \mathrm{Gal}(\bar{\kappa}_A/\kappa_{A_g})$. Hence

$$\partial_A(B)(g) = \mathrm{Res}_{\kappa_A/\kappa_{A_g}}\big(\partial_A(B)\big)(g) = \partial_{A_g}\big(\mathrm{Res}_{k/K^g}(B)\big)(g). \tag{7}$$

Let

$$G = \Gamma_k, \qquad U = \Gamma_{K^g} \quad \text{and} \quad H = \Gamma_K.$$

Choose a system of representatives σ of the double coset decomposition

$$G = \bigsqcup_{\sigma} U\sigma H = \bigsqcup_{\sigma} U\sigma, \tag{8}$$

since H is normal in G. By [10, Proposition I.1.5.6], we have

$$\mathrm{Res}_{G/U} \circ \mathrm{Cor}_{H/G} = \sum_{\sigma} \mathrm{Cor}_{U\cap\sigma H\sigma^{-1}/U} \circ \sigma_* \circ \mathrm{Res}_{H/H\cap\sigma U\sigma^{-1}},$$

where σ runs through all elements in (8). Note that H is a normal subgroup of G and contained in U. Therefore, we have

$$\mathrm{Res}_{G/U} \circ \mathrm{Cor}_{H/G} = \sum_{\sigma} \mathrm{Cor}_{H/U} \circ \sigma_*.$$

Therefore

$$\begin{aligned} \mathrm{Res}_{k/K^g}(B) &= \mathrm{Res}_{G/U} \circ \mathrm{Cor}_{H/G}(\Xi, \chi) = \sum_{\sigma} \mathrm{Cor}_{H/U} \circ \sigma_*(\Xi, \chi) \\ &= \sum_{\sigma} \mathrm{Cor}_{H/U}\big(\sigma(\Xi), \chi^{\sigma}\big), \end{aligned}$$

where χ^{σ} is the conjugate character of χ by σ, *i.e.*, $\chi^{\sigma}(g) := \chi(\sigma^{-1}g\sigma) \in \mathbb{Q}/\mathbb{Z}$ for any $g \in \Gamma_K$.

Since $\mathrm{Gal}(L/K)$ is contained in the center of $\mathrm{Gal}(L/k)$ by Lemma 4, we have

$$\chi^{\sigma} = \chi.$$

Hence

$$\mathrm{Res}_{k/K^g}(B) = \sum_{\sigma} \mathrm{Cor}_{H/U}\big(\sigma(\Xi), \chi\big).$$

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Note that K/K^g is cyclic, we have $\mathrm{Gal}(L/K^g)$ is abelian by the condition $(*)$ over k. Therefore, we can choose a character $\hat{\chi}$ of $\mathrm{Gal}(\bar{k}/K^g)$ which factors through $\mathrm{Gal}(L/K^g)$ and lifts χ. Since σ runs through a system of representatives of $U\backslash G$, we have

$$\begin{aligned}\mathrm{Res}_{k/K^g}(B) &= \sum_{\sigma} \mathrm{Cor}_{H/U}\big(\sigma(\Xi), \mathrm{Res}_{U/H}(\hat{\chi})\big) = \sum_{\sigma} \big(N_{K/K^g}\big(\sigma(\Xi)\big), \hat{\chi}\big) \\ &= \big(N_{K/k}(\Xi), \hat{\chi}\big) = (\alpha, \hat{\chi}).\end{aligned}$$

Obviously

$$\partial_{A_g}\big(\mathrm{Res}_{k/K^g}(B)\big) = v_{A_g}(\alpha)\hat{\chi} = 0.$$

By (7), then $\partial_A(B)(g) = 0$ for any $g \in \mathrm{Gal}(\bar{\kappa}_A/\kappa_A)$. Therefore $B \in \mathrm{Br}(X^c)$.

On the other hand, suppose χ does not factor through any abelian extension over K which satisfies the condition $(*)$ over k. In the following we will show that $B = \mathrm{Cor}_{K/k}(\Xi, \chi) \in \mathrm{Br}(X)$ does not lies in $\mathrm{Br}(X^c)$.

Let L/K be the minimal abelian (cyclic) extension which χ factors through, which is the fixed field by the kernel of $\chi : \Gamma_K \to \mathbb{Q}/\mathbb{Z}$. Then there exists a subfield $F \subset K$ with $\mathrm{Gal}(K/F)$ cyclic, such that L/F is not abelian. Let $f : \mathrm{Br}(X) \to \mathrm{Br}(X_F)$ be the natural map. Let $U' = \Gamma_F$. Choose a system of representatives σ' of the double coset decomposition

$$G = \bigsqcup_{\sigma} U'\sigma' H. \tag{9}$$

By [10, Proposition I.1.5.6], similarly as above we have

$$f(B) = \mathrm{Res}_{G/U'} \circ \mathrm{Cor}_{H/G}(\Xi, \chi) = \sum_{\sigma'} \mathrm{Cor}_{H/U'}\big(\sigma'(\Xi), \chi^{\sigma'}\big).$$

It's clear that X_F is the affine variety defined by

$$\prod_{\sigma'} N_{K/F}\big(\sigma'(\Xi)\big) = \alpha,$$

where σ' runs through all elements in Eq. (9) and all $\sigma'(\Xi)$ are independent K-variables. Let X' be the affine variety over F defined by $N_{K/F}(\Xi') = \alpha$. There is a closed immersion $\varphi : X' \to X_F$ defined by

$$\Xi' \hookleftarrow \Xi, \quad \text{otherwise} \quad 1 \hookleftarrow \sigma'(\Xi) \text{ if } \sigma' \notin \Gamma_F.$$

Hence

$$\varphi^* f(B) = \sum_{\sigma} \varphi^*\big(\mathrm{Cor}_{H/U'}\big(\sigma'(\Xi), \chi^{\sigma'}\big)\big) = \mathrm{Cor}_{H/U'}\big(\Xi', \chi\big),$$

where $\varphi^* : \mathrm{Br}(X_F) \to \mathrm{Br}(X')$ is induced by φ.

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We claim that χ is not the restriction of a character of Γ_F. Otherwise, suppose χ is the restriction of $\chi' : \Gamma_F \to \mathbb{Q}/\mathbb{Z}$. Let L' be the cyclic field fixed by $\mathrm{Ker}(\chi')$. Since $\mathrm{Ker}(\chi) = \mathrm{Ker}(\chi') \cap \Gamma_K$, we have $L = L'.K$. Since L' and K are abelian (cyclic) over F, then $L = L'.K$ is also abelian over F, which is impossible since L/F is not abelian. Therefore, by Proposition 3, we have $\varphi^* f(B) \neq 0 \in \mathrm{Br}(X')/\mathrm{Br}_0(X')$. On the other hand, since K/F is cyclic, it implies $\mathrm{Br}(X'^c) = \mathrm{Br}_0(X'^c)$. Therefore $\varphi^* f(B) \notin \mathrm{Br}(X'^c)$, hence $B \notin \mathrm{Br}(X^c)$. □

Remark 6. Let $K_1, \ldots, K_m$ be finite field extensions of k and $K = \bigcap_{i=1}^m K_i$. Let Y be the variety over k defined by $N_{K_1/k}(\Xi_1) \cdots N_{K_m/k}(\Xi_m) = \alpha$, which is a principle homogeneous space of the multinorm one torus associated with $K_1, \ldots, K_m$. Let X be the variety over k defined by $N_{K/k}(\Xi) = \alpha$. With some special assumptions, there is a canonical and explicit isomorphism

$$\mathrm{Br}\big(X^c\big)/\mathrm{Br}_0\big(X^c\big) \xrightarrow{\simeq} \mathrm{Br}\big(Y^c\big)/\mathrm{Br}_0\big(Y^c\big)$$

by [7, Theorem 6]. Therefore we can also construct $\mathrm{Br}(Y^c)$ using Theorem 5.

Let k be a field with characteristic 0. Let $K = k(\sqrt{-1}, \sqrt{d})$ be a biquadratic extension of k. Let $L = K(\sqrt[4]{d})$. It is clear that L/k is Galois (non-abelian) and of degree 8 with $\mathrm{Gal}(L/k) \cong \mathbf{D}_4$. Let χ be the unique nontrivial character of Γ_K which factors through $\mathrm{Gal}(L/K)$.

Corollary 7. *Let k be a field with characteristic 0. Let $K = k(\sqrt{-1}, \sqrt{d})$ be a biquadratic extension of k. Then $\mathrm{Cor}_{K/k}(\Xi, \chi)$ is the unique generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$.*

Proof. Since $char(k) = 0$, we have $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) \cong Ш^2_\omega(\widehat{T})$ by [4, Proposition 9.5] or [3]. Since K/k is Galois, we have $Ш^2_\omega(\widehat{T}) \cong H^3(\mathrm{Gal}(K/k), \mathbb{Z})$ by a similar argument as in [11, Theorem 6.11]. Therefore we have

$$\mathrm{Br}\big(X^c\big)/\mathrm{Br}_0\big(X^c\big) \cong H^3\big(\mathrm{Gal}(K/k), \mathbb{Z}\big) \cong \mathbb{Z}/2\mathbb{Z}$$

by a classical computation of Schur (see for instance [8, Corollary 2.2.12]).

By Theorem 5 and the fact that each group of order 4 is abelian, we have $\mathrm{Cor}_{K/k}(\Xi, \chi) \in \mathrm{Br}(X^c)$. So it remains to show that $\mathrm{Cor}_{K/k}(\Xi, \chi)$ is nontrivial in $\mathrm{Br}(X^c)$. By Proposition 3, we only need to show that χ is not the restriction of a character of Γ_k. Otherwise, suppose χ is the restriction of $\chi' : \Gamma_k \to \mathbb{Q}/\mathbb{Z}$. Let L' be the cyclic field fixed by $\mathrm{Ker}(\chi')$. Since $\mathrm{Ker}(\chi) = \mathrm{Ker}(\chi') \cap \Gamma_K$, we have $L = L'.K$. Since L' and K are abelian over k, it implies L is also abelian over k, which is impossible since $\mathrm{Gal}(L/k) = \mathbf{D}_4$ is not abelian. □

We will use Corollary 7 to give an explicit example.

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Example 8. For any positive rational number α, we can write $\alpha = 2^{s_1}17^{s_2} \cdot p_1^{e_1} \cdots p_g^{e_g}$, where $p_1, \ldots, p_g$ are distinct primes which do not divide 34, and s_1 and s_2 are integers, and $e_1, \ldots, e_g$ are nonzero integers. Let $\alpha_1 = p_1^{e_1} \cdots p_g^{e_g}$. Let $D(\alpha) = \{p_1, \ldots, p_g\}$. Denote

$$D_1 = \left\{ p \in D(\alpha) \colon \left(\frac{17}{p}\right) = \left(\frac{-17}{p}\right) = -1 \right\}$$

$$D_2 = \left\{ p \in D(\alpha) \colon \left(\frac{17}{p}\right) = \left(\frac{-1}{p}\right) = 1 \text{ and } \left(\frac{17}{p}\right)_4 = -1 \right\},$$

where $(\frac{\cdot}{\cdot})$ is the Legendre symbol and $(\frac{\cdot}{\cdot})_4$ is the quartic residue symbol. Let $K = \mathbb{Q}(\sqrt{-1}, \sqrt{17})$. Then the equation

$$N_{K/\mathbb{Q}}(\Xi) = \alpha \tag{10}$$

is solvable over $\mathbb{Q}$ if and only if the following conditions hold:

(i) (locally solvable): $(\frac{\alpha_1, -1}{2}) = (\frac{\alpha_1}{17}) = 1$, where $(\frac{\cdot,\cdot}{2})$ is the Hilbert symbol over $\mathbb{Q}_2$; e_i is even if $(\frac{-1}{p_i}) = -1$ or $(\frac{17}{p_i}) = -1$.
(ii) (BM-obstruction): $(-1)^{s_1 + \sum_{p_i \in D_1} e_i/2 + \sum_{p_i \in D_2} e_i} = (\frac{\alpha_1}{17})_4$.

Proof. First we consider the local solvability of (10). It is clear that (10) is solvable at the infinite place and any finite place $p \nmid 34\alpha$, and (10) is solvable at place 2 if and only if $(\frac{\alpha_1, -1}{2}) = 1$, and (10) is solvable at place 17 if and only if $(\frac{\alpha_1}{17}) = 1$. If $p = p_i$ with $(\frac{-1}{p_i}) = (\frac{17}{p_i}) = 1$, then $K/\mathbb{Q}$ is totally split at p, then (10) is solvable at p. If $p = p_i$ with $(\frac{-1}{p_i}) = -1$ or $(\frac{17}{p_i}) = -1$, then $K \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is a product of two unramified extensions over $\mathbb{Q}_p$ of degree 2, hence (10) is solvable at place p_i if and only if e_i is even.

Now we consider the Brauer–Manin obstruction. We assume (10) is solvable at all places. Let $L = K(\sqrt[4]{17})$. Then $B = \mathrm{Cor}_{K/k}(\Xi, \chi)$ is the unique generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ by Corollary 7, where χ be the unique nontrivial character of Γ_K which factors through $\mathrm{Gal}(L/K)$. Let p be a prime number (or $p = \infty$) and let $x_p = (\Xi_p) \in X(\mathbb{Q}_p)$ with $\Xi_p \in (K \otimes \mathbb{Q}_p)^*$.

If $p = \infty$, then χ_p is trivial since K is totally imaginary, hence $B(x_p) = 0$.

Let p be a finite prime, we fix an embedding $\bar{\mathbb{Q}} \hookrightarrow \bar{\mathbb{Q}}_p$ and let $K' = \mathbb{Q}_p \cap K$. We can see K' has the following cases:

- if $p = 2$, then $K' = \mathbb{Q}(\sqrt{7})$.
- if $p = 17$, then $K' = \mathbb{Q}(\sqrt{-1})$.
- if $p \nmid 34$, then $K/\mathbb{Q}$ is unramified at p. Since $\mathrm{Gal}(K/\mathbb{Q}) = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, K' contains a biquadratic fields.

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Therefore, we have $2 \mid [K' : \mathbb{Q}]$ for all $p < \infty$. By a similar arguments as in the proof of Theorem 5, in the following we will show that the restriction $\mathrm{Res}_{\mathbb{Q}/K'}(B)$ of B to $\mathrm{Br}(X \times_{\mathbb{Q}} K')$ is the constant (α, χ'), where χ' is a character of $\Gamma_{K'}$ which lifts χ.

Using the notation in [10, Proposition I.1.5.6], let $G = \Gamma_{\mathbb{Q}}, U = \Gamma_{K'}, H = \Gamma_K \subset U$, we have the following decomposition

$$G = \bigsqcup_{\sigma} U\sigma H = \bigsqcup_{\sigma} U\sigma,$$

where σ runs through a finite system of representatives of the double cosets.

Then we have

$$\begin{aligned}\mathrm{Res}_{\mathbb{Q}/K'}(B) &= \mathrm{Res}_{G/U} \circ \mathrm{Cor}_{H/G}(\Xi, \chi) = \sum_{\sigma} \mathrm{Cor}_{U/H}\big(\sigma(\Xi), \chi^{\sigma}\big) \\ &= \sum_{\sigma} \mathrm{Cor}_{U/H}\big(\sigma(\Xi), \chi\big),\end{aligned}$$

where the last equation holds since $\chi = \chi^{\sigma}$ (note that χ has order 2). Since $2 \mid [K' : \mathbb{Q}]$ and $\mathrm{Gal}(L/\mathbb{Q}) = \mathbf{D}_4$, hence $\mathrm{Gal}(L/K')$ is abelian. Therefore we can lift χ to be a character χ' of $\Gamma_{K'}$ factoring through $\mathrm{Gal}(L/K')$ and we have

$$\begin{aligned}\mathrm{Res}_{\mathbb{Q}/K}(B) &= \sum_{\sigma} \mathrm{Cor}_{V/U}\big(\sigma(\Xi), \mathrm{Res}_{U/V}(\chi')\big) = \sum_{\sigma} \big(N_{K/K'}\big(\sigma(\Xi)\big), \chi'\big) \\ &= \big(N_{L/\mathbb{Q}}(\Xi), \chi'\big) = \big(\alpha, \chi'\big). \qquad (11)\end{aligned}$$

Hence we have $B(x_p) = 0$ for $p \neq \infty$ and $p \nmid 34\alpha$.

If $p = 2$, then $K' = \mathbb{Q}(\sqrt{17})$. We can choose χ' to be the generator of $H^1(K'(\sqrt[4]{17})/K', \mathbb{Q}/\mathbb{Z}) \subset H^1(K', \mathbb{Q}/\mathbb{Z})$. However $x^4 = 17$ is solvable in $\mathbb{Q}_2$. Then we have $B(x_2) = (\alpha, \chi')_2 = 0$.

If $p = 17$, then $K' = \mathbb{Q}(\sqrt{-1})$. We can choose χ' to be the generator of $H^1(K'(\sqrt[4]{17})/K', \mathbb{Q}/\mathbb{Z}) \subset H^1(K', \mathbb{Q}/\mathbb{Z})$. Then we have $B(x_{17}) = (\alpha, \chi')_{17} = (-1)^{s_1} \times (\frac{\alpha_1}{17})_4$.

If $p = p_i$ with $(\frac{17}{p_i}) = 1$, then $K' \supset \mathbb{Q}(\sqrt{17})$. We can choose χ' to be the generator of $H^1(K'(\sqrt[4]{17})/K', \mathbb{Q}/\mathbb{Z}) \subset H^1(K', \mathbb{Q}/\mathbb{Z})$, it has degree 2. Then

$$B(x_{p_i}) = \big(\alpha, \chi'\big)_{p_i} = \left(\frac{\alpha, \sqrt{17}}{p_i}\right) = \left(\frac{17}{p_i}\right)_4^{e_i}.$$

We will discuss it by the following cases:

- if $(\frac{-1}{p_i}) = -1$, then e_i is even by local solvability, hence $B(x_{p_i}) = (\frac{17}{p_i})_4^{e_i} = (\frac{17}{p_i})^{e_i/2} = 1$.
- if $(\frac{-1}{p_i}) = 1$ and $(\frac{17}{p})_4 = 1$, obviously $B(x_{p_i}) = 1$.

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- if $(\frac{-1}{p_i}) = 1$ and $(\frac{17}{p})_4 = -1$, obviously $B(x_{p_i}) = (\frac{17}{p_i})_4^{e_i} = (-1)^{e_i}$.

If $p = p_i$ with $(\frac{17}{p_i}) = -1$, then e_i is even by local solvability. We have $(\frac{-1}{p_i}) = 1$ or $(\frac{-17}{p_i}) = 1$. We will discuss it by the following cases:

- if $(\frac{-1}{p_i}) = 1$, then $K' = \mathbb{Q}(\sqrt{-1})$. We can choose χ' to be a generator of $H^1(K'(\sqrt[4]{17})/K', \mathbb{Q}/\mathbb{Z}) \subset H^1(K', \mathbb{Q}/\mathbb{Z})$, it is of degree 4. Hence $B(x_{p_i}) = (\frac{17}{p_i})_4^{e_i} = (\frac{17}{p_i})^{e_i/2} = (-1)^{e_i/2}$.
- if $(\frac{-17}{p_i}) = 1$, then $K' = \mathbb{Q}(\sqrt{-17})$. Since $\sqrt{17} = \sqrt{-1}\cdot\sqrt{-17} = 2\sqrt{-17}\cdot(1+\sqrt{-1})^2/4$, we have $L = K'(\sqrt{-1}, \sqrt{2\sqrt{-17}})$. Therefore we can choose χ' to be contained in $H^1(L/K', \mathbb{Q}/\mathbb{Z}) \subset H^1(K', \mathbb{Q}/\mathbb{Z})$, it is of degree 1 or 2. Since e_i is even, we have $B(x_{p_i}) = (\alpha, \chi')_{p_i} = (p_i^{e_i}, \chi')_{p_i} = 1$.

By the above computation, we have

$$B(x_p) = \begin{cases} (-1)^{s_1}(\frac{\alpha_1}{17})_4 & \text{if } p = 17 \\ (-1)^{e_i/2} & \text{if } p = p_i \in D_1 \\ (-1)^{e_i} & \text{if } p = p_i \in D_2 \\ 1 & \text{otherwise.} \end{cases}$$

The proof follows from [13, Theorem 8.12] or [14, Chapter 5]. □

Let k be a field with characteristic 0. Let $K = k(\sqrt{d_1}, \sqrt{d_2})$ be a biquadratic extension of k. Suppose $x^2 - d_1y^2 = d_2z^2$ has a nonzero solution (x_0, y_0, z_0) in k. Let $L = K(\sqrt{x_0 + y_0\sqrt{d_1}})$. We can see L/k is Galois (non-abelian) and of degree 8 with $\mathrm{Gal}(L/k) \cong \mathbf{D}_4$. Let χ be the unique nontrivial character of $\mathrm{Gal}(\bar{k}/K)$ which factors through $\mathrm{Gal}(L/K)$. Similarly as above we immediately have the following result:

Corollary 9. *Let k be a field with characteristic 0. Let $K = k(\sqrt{d_1}, \sqrt{d_2})$ be as above. Then $\mathrm{Cor}_{K/k}(\Xi, \chi)$ is the unique generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$.*

Finally we will use Corollary 9 to give an explicit example.

Example 10. For any nonzero rational number α, we can write $\alpha = (-1)^{s_0}2^{s_1}13^{s_2}17^{s_3} \times p_1^{e_1}\cdots p_g^{e_g}$, where $p_1, \ldots, p_g$ are distinct primes which do not divide 442, and $s_0, \ldots, s_2$ are integers, and $e_1, \ldots, e_g$ are nonzero integers. Let $D(\alpha) = \{p_1, \ldots, p_g\}$ and $\alpha_1 = p_1^{e_1}\cdots p_g^{e_g}$. Let $K = \mathbb{Q}(\sqrt{13}, \sqrt{17})$. Denote

$$D_1 = \left\{p \in D(\alpha):\ \left(\frac{13}{p}\right) = \left(\frac{17}{p}\right) = -1\right\}$$

$$D_2 = \left\{p \in D(\alpha):\ \left(\frac{13}{p}\right) = \left(\frac{17}{p}\right) = 1 \text{ and } \left(\frac{15 + 4\sqrt{13}}{p}\right) = -1\right\}.$$

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Let $K = \mathbb{Q}(\sqrt{13}, \sqrt{17})$. Then the equation $N_{K/\mathbb{Q}}(\Xi) = \alpha$ is solvable over $\mathbb{Q}$ if and only if the following conditions hold:

(i) (locally solvable): s_1 is even; and $(\frac{\alpha_1}{13}) = (\frac{\alpha_1}{17}) = 1$; and e_i is even if $(\frac{13}{p_i}) = -1$ or $(\frac{17}{p_i}) = -1$.
(ii) (BM-obstruction): $(-1)^{\sum_{p_i \in D_1} e_i/2 + \sum_{p_i \in D_2} e_i} = (-1)^{s_0+s_2} \cdot (\frac{\alpha_1, -1}{2})$, where $(\frac{\cdot,\cdot}{2})$ is the Hilbert symbol over $\mathbb{Q}_2$.

Proof. We can see $15^2 - 13 \cdot 4^2 = 17$. Let $L = K(\sqrt{15 + 4\sqrt{13}})$. Then $B = \mathrm{Cor}_{K/k}(\Xi, \chi)$ is the unique generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ by Corollary 7, where χ is the unique nontrivial character of Γ_K which factors through $\mathrm{Gal}(L/K)$. The proof follows from a similar argument as in Example 8. □

Remark 11.

(a) By the Poitou–Tate duality, for any torus T, if the Tate–Shafarevich group $Ш^2(\widehat{T})$ is trivial, then the Hasse principle holds on any principal homogeneous space of T. For the two examples as above, since $Ш^2(\widehat{T}) = \mathbb{Z}/2\mathbb{Z}$ is non-trivial, the Hasse principle does not hold.
(b) Let k be a number field and $K = k(\sqrt{a}, \sqrt{b})$ a biquadratic extension over k. Sansuc [13, Proposition 6] gave a method to determine the solvability of the equation $N_{K/k}(\Xi) = \alpha \in k^\times$. For each $\alpha \in k^\times$, his method needs to look for a rational solution $(\Xi_1, \Xi_2) \in k(\sqrt{a})^\times \times k(\sqrt{b})^\times$ of the equation

$$N_{k(\sqrt{a})/k}(\Xi_1) \cdot N_{k(\sqrt{b})/k}(\Xi_2) = \alpha.$$

Generally, looking for a rational point is not easy. By our method, the unramified Brauer groups for all $\alpha \in k^\times$ have the same form, then the solvability for any α can be uniformly tested, e.g., we only need to compute some Legendre or Hilbert symbols for all α in Examples 8 and 10. On the other hand, our method can also work on any Galois extension (not only biquadratic), see Example 15 for a cyclotomic case.

2. The case that $k = \mathbb{Q}$ and $K/\mathbb{Q}$ is biquadratic

In Section 2.1, we will recall some results of double coverings of $\mathbb{Q}^{ab}/\mathbb{Q}$. In Section 2.2, an explicit construction for the biquadratic case will be given using Theorem 5 in Section 1 and the double coverings in Section 2.1.

2.1. Double coverings of $\mathbb{Q}^{ab}/\mathbb{Q}$

Suppose K/F is Galois. A double covering of K/F (defined in [6]) is an extension $\hat{K}/K$ of degree $\leqslant 2$ such that $\hat{K}/F$ is Galois. Let $\mathbb{Q}^{ab}$ be the maximal abelian extension of $\mathbb{Q}$.

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In the following we will describe all double coverings of $\mathbb{Q}^{ab}/\mathbb{Q}$ (see [1,6]) and of the cyclotomic extension $\mathbb{Q}(\xi_n)/\mathbb{Q}$ (see [15]).

Let $\mathscr{A}$ be the free abelian group on the symbols of the form $[a]$ $(a \in \mathbb{Q})$ modulo the identifications

$$[a] = [b] \quad \Leftrightarrow \quad a - b \in \mathbb{Z}.$$

For all odd primes $p < q$, define $\mathbf{a}_{pq} \in \mathscr{A}$ as

$$\mathbf{a}_{pq} = \sum_{i=1}^{\frac{p-1}{2}} \left(\left[\frac{i}{p}\right] - \sum_{k=0}^{\frac{q-1}{2}} \left[\frac{i}{pq} + \frac{k}{q}\right] \right) - \sum_{j=1}^{\frac{q-1}{2}} \left(\left[\frac{j}{q}\right] - \sum_{l=0}^{\frac{p-1}{2}} \left[\frac{j}{pq} + \frac{l}{p}\right] \right)$$

and for prime $q > 2$, define $\mathbf{a}_{2q} \in \mathscr{A}$ as

$$\mathbf{a}_{2q} = \left(\left[\frac{1}{4}\right] - \sum_{k=0}^{\frac{q-1}{2}} \left[\frac{k}{q} + \frac{1}{4q}\right] \right) - \sum_{j=1}^{\frac{q-1}{2}} \left(\left[\frac{j}{q}\right] + \left[\frac{j}{q} - \frac{1}{2q}\right] - \left[\frac{j}{2q}\right] - \left[\frac{j}{2q} - \frac{1}{4q}\right] \right).$$

Let

$$\sin : \mathscr{A} \to {\mathbb{Q}^{ab}}^{\times}$$

be the unique homomorphism such that

$$\sin[a] = \begin{cases} 2\sin(\pi a)\ (= |1 - e^{2\pi i a}|) & \text{if } 0 < a < 1 \\ 1 & \text{if } a = 0 \end{cases} \qquad \big(a \in \mathbb{Q} \cap [0,1)\big).$$

The composite field of all double coverings of $\mathbb{Q}^{ab}/\mathbb{Q}$ is (see [1, main theorem])

$$\mathbb{Q}^{ab}\Big(\big\{ \sqrt[4]{l} \big\}_{l:\text{prime}} \cup \big\{ \sqrt{\sin \mathbf{a}_{pq}} \big\}_{\substack{p,q:\text{prime} \\ p<q}} \Big).$$

Now we come to the cyclotomic case. First we give some notation. Let $n \not\equiv 2 \bmod 4$, we define a subset S_n of $\mathbb{Z}$ associated to n as following:

i) if $2 \nmid n$, set $S_n := \{\text{odd prime factors of } n\}$;
ii) if $4 \mid n$ and $8 \nmid n$, set $S_n := \{-1\} \cup \{\text{odd prime factors of } n\}$;
iii) if $8 \mid n$, set $S_n := \{-1, 2\} \cup \{\text{odd prime factors of } n\}$.

If $4 \mid n$, then for all $p, q \in S_n$ and $p < q$, we set

$$u_{pq} := \begin{cases} \sqrt{q} & \text{if } p = -1 \\ \sin \mathbf{a}_{pq} & \text{otherwise.} \end{cases}$$

If $2 \nmid n$, then for primes $p, q \in S_n$ and $p < q$, we set

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$$u_{pq} := \begin{cases} \sin \mathbf{a}_{pq} & \text{if } p \equiv q \equiv 1 \bmod 4 \\ \sqrt{p} \cdot \sin \mathbf{a}_{pq} & \text{if } p \equiv 1, q \equiv 3 \bmod 4 \\ \sqrt{q} \cdot \sin \mathbf{a}_{pq} & \text{if } p \equiv 3, q \equiv 1 \bmod 4 \\ \sqrt{pq} \cdot \sin \mathbf{a}_{pq} & \text{if } p \equiv q \equiv 3 \bmod 4, \end{cases} \tag{12}$$

Let $K = \mathbb{Q}(\xi_n)$, where ξ_n is a primitive root of unity. Then the composite field of all double coverings of $K/\mathbb{Q}$ is (see [15, Theorem 1])

$$K\big(\{\sqrt{u_{pq}}\}_{p<q\in S_n}\big).K',$$

where $K' = K(\{\sqrt{-1}\} \cup \{\sqrt{l}\}_{l:\text{prime}})$.

2.2. Construction of the Brauer group

Let $K = \mathbb{Q}(\sqrt{d_1 d_2}, \sqrt{d_1 d_3})$ such that $d_1, d_2, d_3 \in \mathbb{Z}$ are square-free and relatively prime to each other. Without loss generality, we can assume $d_1 d_2 > 0$. In this section, we will explicitly construct the unramified Brauer group of the affine variety X over $\mathbb{Q}$ defined by $N_{K/k}(\Xi) = \alpha$, where $\alpha \in \mathbb{Q}^\times$ fixed.

Denote

$$\begin{aligned} S_i &= \{p \text{ rational prime: } p \mid d_i\} \quad \text{for } 1 \leqslant i \leqslant 3, \\ R &= \bigcup_{1\leqslant i<j\leqslant 3} S_i \times S_j, \\ N &= \begin{cases} |d_1 d_2 d_3| & \text{if } d_1 d_2 \equiv d_1 d_3 \equiv 1 \bmod 4 \\ 4|d_1 d_2 d_3| & \text{otherwise.} \end{cases} \end{aligned} \tag{13}$$

Let $F = \mathbb{Q}(\xi_N)$. It is clear that K is contained in the cyclotomic field F. For simplicity of the notation, we extend the definition of $\mathbf{a}_{pq}$ and u_{pq} for $p > q$ by

$$\mathbf{a}_{pq} = \mathbf{a}_{qp} \quad \text{and} \quad u_{pq} = u_{qp}.$$

Let

$$\begin{aligned} \Delta &= \begin{cases} \prod_{(p,q)\in R} \sin \mathbf{a}_{pq} & \text{if } d_1 d_3 > 0 \\ \sqrt{d_1 d_2} \prod_{(p,q)\in R} \sin \mathbf{a}_{pq} & \text{if } d_1 d_3 < 0, \end{cases} \\ L &= F(\sqrt{\Delta}). \end{aligned}$$

We will see the generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ is constructed by a character associated to L/K.

Lemma 12. *The field extension $L/\mathbb{Q}$ is Galois and $L \not\subset \mathbb{Q}^{ab}$.*

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Proof. If $4 \mid N$, it follows from Theorems 11 and 12 in [6]. So we only need to consider the case $4 \nmid N$, *i.e.*, $d_1 d_2 \equiv d_1 d_3 \equiv 1 \bmod 4$.

Let $\Delta' = \prod_{(p,q)\in R} u_{pq}$. By an easy computation, we have

$$\Delta' = \prod_{(p,q)\in R} \sin \mathbf{a}_{pq} \prod_{p \mid d_1 d_2 d_3} \sqrt{p^{e_p}},$$

where

$$e_p = \begin{cases} \#\{q \mid d_2 d_3 \colon\ q \equiv 3 \bmod 4\} & \text{if } p \mid d_1 \\ \#\{q \mid d_1 d_3 \colon\ q \equiv 3 \bmod 4\} & \text{if } p \mid d_2 \\ \#\{q \mid d_1 d_2 \colon\ q \equiv 3 \bmod 4\} & \text{if } p \mid d_3. \end{cases}$$

Since $d_1 d_2 > 0$, it implies e_p is even when $p \mid d_3$.

If $d_1 d_3 > 0$, we also have $d_2 d_3 > 0$. Then e_p is even when $p \mid d_1 d_2$. So $\Delta' = \pm\Delta \cdot u^2$ with $u \in F^\times$. Therefore $L = F(\sqrt{\Delta}) = F(\sqrt{\Delta'})$ or $F(\sqrt{-\Delta'})$. Then $L/\mathbb{Q}$ is Galois and $L \not\subset \mathbb{Q}^{ab}$ by [15, Theorem 1 and Proposition 1].

If $d_1 d_3 < 0$, we also have $d_2 d_3 < 0$. Then e_p is odd when $p \mid d_1 d_2$. So $\Delta' = \pm\Delta \cdot u^2$ with $u \in F^\times$. Therefore $L = F(\sqrt{\Delta}) = F(\sqrt{\Delta'})$ or $F(\sqrt{-\Delta'})$. Then $L/\mathbb{Q}$ is Galois and $L \not\subset \mathbb{Q}^{ab}$ by [15, Theorem 1 and Proposition 1]. □

Theorem 13. *Let X be the affine variety over $\mathbb{Q}$ defined by $N_{K/\mathbb{Q}}(\Xi) = \alpha \in \mathbb{Q}^\times$. There is a character χ of Γ_K which factors through $\mathrm{Gal}(L/K)$ and nontrivial on $\mathrm{Gal}(L/F)$. Such χ gives an element $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi, \chi)$ in $\mathrm{Br}(X^c)$ which generates $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$.*

Proof. First we will show that L/K satisfies the condition $(*)$ over $\mathbb{Q}$, hence L/K is abelian, then there exists a character χ of Γ_K which factors through $\mathrm{Gal}(L/K)$ and nontrivial on $\mathrm{Gal}(L/F)$. By Theorem 5, we have $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi, \chi) \in \mathrm{Br}(X^c)$.

Let K' be a subfield of K such that K/K' is cyclic. We want to show $\mathrm{Gal}(L/K')$ is abelian. The extension $\mathbb{Q}^{ab}(\sqrt{\Delta})/K'$ is Galois by the main theorem in [1]. Since $\mathrm{Gal}(L/K')$ is a quotient of $\mathrm{Gal}(\mathbb{Q}^{ab}(\sqrt{\Delta})/K')$, we only need to show $\mathrm{Gal}(\mathbb{Q}^{ab}(\sqrt{\Delta})/K')$ is abelian. Let $G^{ab} = \mathrm{Gal}(\mathbb{Q}^{ab}/\mathbb{Q})$. There is the central extension

$$\Sigma : 0 \to \mathbb{Z}/2\mathbb{Z} \to \mathrm{Gal}\big(\mathbb{Q}^{ab}(\sqrt{\Delta})/\mathbb{Q}\big) \to G^{ab} \to 0.$$

Let

$$\vartheta : G^{ab} \to \mathrm{Gal}\big(\mathbb{Q}^{ab}(\sqrt{\Delta})/\mathbb{Q}\big)$$

be any set-theoretic splitting of the central extension Σ. Then Σ gives a 2-cocycle

$$a \in Z^2\big(G^{ab}, \mathbb{Z}/2\mathbb{Z}\big)$$

defined by the formula

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$$a_{\sigma,\tau} = \vartheta(\sigma)\vartheta(\tau)\vartheta(\sigma\tau)^{-1} \quad \big(\sigma,\tau \in G^{ab}\big).$$

Let $\beta \in Z^2(G^{ab}, \mathbb{Z}/2\mathbb{Z})$ defined by

$$\beta_{\sigma,\tau} = a_{\sigma,\tau} - a_{\tau,\sigma} \quad \big(\sigma,\tau \in \mathrm{Gal}\big(\mathbb{Q}^{ab}/K'\big)\big).$$

It is easy to check that β is a skew-symmetric (symmetric) bilinear map.

For each odd prime p, let $G_p \subset G^{ab}$ be the inertia subgroup at p, which is isomorphic to $\mathbb{Z}_p^\times$. Let $G_{-1} \subset G^{ab}$ be the subgroup generated by the restriction of complex conjugation to $\mathbb{Q}^{ab}$. Let $G_2 \subset G^{ab}$ be the subgroup of the inertial subgroup at 2 fixing $\sqrt{-1}$, which is isomorphic to $1 + 4\mathbb{Z}_2$. Let $S = \{-1\} \cup \{p \mid p \text{ is rational prime}\}$. By Kronecker–Weber's theorem, we have

$$G^{ab} = \prod_{p\in S} G_p.$$

For $p \in S$ the profinite group G_p is procyclic. Let $\sigma_p \in G^{ab}$ such that σ_p projects to a topological generator of G_p and projects to $1 \in G_q$ for $q \neq p$. For $p \in S$, the natural quotient map $G^{ab} \to \mathrm{Gal}(\mathbb{Q}(\sqrt{p})/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$ induces the map $G_p \to \mathbb{Z}/2\mathbb{Z}$ by $\sigma_p^{i_p} \mapsto \bar{i}_p$.

(1) Suppose $d_1 d_3 > 0$. Then $\Delta = \prod_{(p,q)\in R} \sin \mathbf{a}_{pq}$ by our definition. By the Log wedge Formula in §4.3.4 and Proposition 2.9 in [1], we have

$$\beta = \sum_{(p,q)\in R} \delta_{p,q} \in Z^2\big(G^{ab}, \mathbb{Z}/2\mathbb{Z}\big),$$

where $\delta_{p,q} = \delta_{q,p} : G^{ab} \times G^{ab} \to \mathbb{Z}/2\mathbb{Z}$ is defined by

$$\big(\big(\sigma_l^{i_{l,1}}\big)_{l\in S}, \big(\sigma_l^{i_{l,2}}\big)_{l\in S}\big) \mapsto \bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}$$

with $i_{-1,1}, i_{-1,2} = 0$ or 1 and $i_{l,1}, i_{l,2} \in \mathbb{Z}_l$ for $l \neq -1$, and see (13) for the definition of R.

(2) Suppose $d_1 d_3 < 0$. Then $\Delta = \sqrt{d_1 d_2} \prod_{(p,q)\in R} \sin \mathbf{a}_{pq}$. By the Log wedge Formula in §3.4, §4.3.4 and Proposition 2.9 in [1], we have

$$\beta = \sum_{(p,q)\in R} \delta_{p,q} + \sum_{p \mid d_1 d_2} \delta_{-1,p},$$

where $\delta_{-1,p} : G^{ab} \times G^{ab} \to \mathbb{Z}/2\mathbb{Z}$ is defined by

$$\big(\big(\sigma_l^{i_{l,1}}\big)_{l\in S}, \big(\sigma_l^{i_{l,2}}\big)_{l\in S}\big) \mapsto \bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1},$$

with $i_{-1,1}, i_{-1,2} = 0$ or 1 and $i_{l,1}, i_{l,2} \in \mathbb{Z}_l$ for $l \neq -1$.

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We have the central extension

$$\Sigma_{K'} : 0 \to \mathbb{Z}/2\mathbb{Z} \to \mathrm{Gal}\big(\mathbb{Q}^{ab}(\sqrt{\Delta})/K'\big) \to \mathrm{Gal}\big(\mathbb{Q}^{ab}/K'\big) \to 0.$$

Then $\Sigma_{K'}$ gives a 2-cocycle $a' = \mathrm{Res}_{\mathbb{Q}/K'}(a) \in Z^2(\mathrm{Gal}(\mathbb{Q}^{ab}/K'), \mathbb{Z}/2\mathbb{Z})$. Let

$$\beta'_{\sigma,\tau} = a'_{\sigma,\tau} - a'_{\tau,\sigma} \quad \big(\sigma, \tau \in \mathrm{Gal}\big(\mathbb{Q}^{ab}/K'\big)\big).$$

We can verify (see [1, Lemma 2.8]) that $\mathrm{Gal}(\mathbb{Q}^{ab}(\sqrt{\Delta})/K')$ is abelian if and only if

$$\beta'_{\sigma,\tau} = 0 \quad \text{for any } \sigma, \tau \in \mathrm{Gal}\big(\mathbb{Q}^{ab}/K'\big). \tag{14}$$

In the following we will check β' satisfies (14).

(i) Suppose $d_1 d_3 > 0$. We can assume $K' = \mathbb{Q}(\sqrt{d_1 d_2})$. The other cases are similar since $d_1 d_2, d_1 d_3$ and $d_2 d_3$ are all positive. Let

$$g_1 = \big(\sigma_p^{i_{p,1}}\big)_{p\in S}, \qquad g_2 = \big(\sigma_p^{i_{p,2}}\big)_{p\in S} \in \mathrm{Gal}\big(\mathbb{Q}^{ab}/K'\big) \subset G^{ab}.$$

Since g_1, g_2 fix K', we have

$$\sum_{p|d_1 d_2} \bar{i}_{p,j} = \sum_{p|d_1} \bar{i}_{p,j} + \sum_{p|d_2} \bar{i}_{p,j} = 0 \quad \text{for } j = 1, 2.$$

Therefore

$$\begin{aligned}
\beta'_{g_1,g_2} = \beta_{g_1,g_2} &= \sum_{(p,q)\in R} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) \\
&= \sum_{(p,q)\in S_1\times S_2} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) + \sum_{(p,q)\in S_1\times S_3} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) \\
&\quad + \sum_{(p,q)\in S_2\times S_3} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) \\
&= \sum_{p|d_1} \bar{i}_{p,1} \sum_{p|d_2} \bar{i}_{p,2} + \sum_{p|d_1} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} + \sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} \\
&\quad + \sum_{p|d_1} \bar{i}_{p,2} \sum_{p|d_2} \bar{i}_{p,1} + \sum_{p|d_1} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} + \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} \\
&= -\sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_2} \bar{i}_{p,2} - \sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} + \sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} \\
&\quad - \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_2} \bar{i}_{p,1} - \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} + \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} \\
&= 0.
\end{aligned}$$

(ii) Suppose $d_1 d_3 < 0$.

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(a) Suppose $K' = \mathbb{Q}(\sqrt{d_1 d_2})$. Let

$$g_1 = \left(\sigma_p^{i_{p,1}}\right)_{p\in S}, \qquad g_2 = \left(\sigma_p^{i_{p,2}}\right)_{p\in S} \in \mathrm{Gal}\left(\mathbb{Q}^{ab}/K'\right) \subset G^{ab}.$$

Since g_1, g_2 fix K', we have

$$\sum_{p|d_1 d_2} \bar{i}_{p,j} = 0 \quad \text{for } j = 1, 2.$$

Similar as above one has

$$\begin{aligned}
\beta'_{g_1,g_2} = \beta_{g_1,g_2} &= \sum_{(p,q)\in R} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) + \sum_{p|d_1 d_2} (\bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1}) \\
&= 0 + \bar{i}_{-1,1} \sum_{p|d_1 d_2} \bar{i}_{p,2} + \bar{i}_{-1,2} \sum_{p|d_1 d_2} \bar{i}_{p,1} \\
&= 0.
\end{aligned}$$

(b) Suppose $K' = \mathbb{Q}(\sqrt{d_1 d_3})$ (similar proof for $K' = \mathbb{Q}(\sqrt{d_2 d_3})$). Let

$$g_1 = \left(\sigma_p^{i_{p,1}}\right)_{p\in S}, \qquad g_2 = \left(\sigma_p^{i_{p,2}}\right)_{p\in S} \in \mathrm{Gal}\left(\mathbb{Q}^{ab}/K'\right) \subset G^{ab}.$$

Since g_1, g_2 fix K' and $d_1 d_3 < 0$, we have

$$\bar{i}_{-1,j} + \sum_{p|d_1 d_3} \bar{i}_{p,j} = 0 \quad \text{for } j = 1, 2.$$

Therefore

$$\begin{aligned}
\beta'_{g_1,g_2} = \beta_{g_1,g_2} &= \sum_{(p,q)\in R} (\bar{i}_{p,1}\bar{i}_{q,2} + \bar{i}_{p,2}\bar{i}_{q,1}) + \sum_{p|d_1 d_2} (\bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1}) \\
&= \sum_{p|d_1} \bar{i}_{p,1} \sum_{p|d_2} \bar{i}_{p,2} + \sum_{p|d_1} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} + \sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} \\
&\quad + \sum_{p|d_1} \bar{i}_{p,2} \sum_{p|d_2} \bar{i}_{p,1} + \sum_{p|d_1} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} + \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1} \\
&\quad + \sum_{p|d_1 d_2} (\bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1}) \\
&= -\left(\bar{i}_{-1,1} + \sum_{p|d_3} \bar{i}_{p,1}\right) \sum_{p|d_2} \bar{i}_{p,2} - \left(\bar{i}_{-1,1} + \sum_{p|d_3} \bar{i}_{p,1}\right) \sum_{p|d_3} \bar{i}_{p,2} + \sum_{p|d_2} \bar{i}_{p,1} \sum_{p|d_3} \bar{i}_{p,2} \\
&\quad - \left(\bar{i}_{-1,2} + \sum_{p|d_3} \bar{i}_{p,2}\right) \sum_{p|d_2} \bar{i}_{p,1} - \left(\bar{i}_{-1,2} + \sum_{p|d_3} \bar{i}_{p,2}\right) \sum_{p|d_3} \bar{i}_{p,1} + \sum_{p|d_2} \bar{i}_{p,2} \sum_{p|d_3} \bar{i}_{p,1}
\end{aligned}$$

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$$
\begin{aligned}
&+ \sum_{p|d_1d_2} (\bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1}) \\
&= -\bar{i}_{-1,1} \sum_{p|d_2d_3} \bar{i}_{p,2} - \bar{i}_{-1,2} \sum_{p|d_2d_3} \bar{i}_{p,1} + \sum_{p|d_1d_2} (\bar{i}_{-1,1}\bar{i}_{p,2} + \bar{i}_{-1,2}\bar{i}_{p,1}) \\
&= \bar{i}_{-1,1} \sum_{p|d_1d_3} \bar{i}_{p,2} + \bar{i}_{-1,2} \sum_{p|d_1d_3} \bar{i}_{p,1} \\
&= -\bar{i}_{-1,1}\bar{i}_{-1,2} - \bar{i}_{-1,2}\bar{i}_{-1,1} = 0.
\end{aligned}
$$

Therefore L/K satisfies the condition $(*)$ over $\mathbb{Q}$, hence $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi, \chi) \in \mathrm{Br}(X^c)$.

Since $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) \cong H^3(\mathrm{Gal}(K/\mathbb{Q}), \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, we only need to show $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi, \chi)$ is nontrivial. Recall

$$
F = \mathbb{Q}(\xi_N), \qquad K \subset F \subset \mathbb{Q}^{ab} \quad \text{and} \quad L = F(\sqrt{\Delta}).
$$

By Proposition 3, we only need to show that χ is not the restriction of a character of $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$. We assume that χ is the restriction of a character $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$, hence χ is trivial on the closure of the commutator subgroup of $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$ which is just $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}^{ab})$. Then χ is trivial on $\mathrm{Gal}(\bar{\mathbb{Q}}/\mathbb{Q}^{ab})$. On the other hand, χ factors through $\mathrm{Gal}(L/K)$. Therefore we have χ is trivial on $\mathrm{Gal}(\bar{\mathbb{Q}}/L \cap \mathbb{Q}^{ab})$. Note that $L/\mathbb{Q}$ is non-abelian (see Lemma 12), hence $F = L \cap \mathbb{Q}^{ab}$. Therefore χ is trivial on $\mathrm{Gal}(\bar{\mathbb{Q}}/F)$, which is impossible since χ factors through $\mathrm{Gal}(L/K)$ and is nontrivial on $\mathrm{Gal}(L/F)$. □

For the cyclotomic case, the 2-torsion part of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ can also be given by Theorem 5.

Proposition 14. *Let $m \not\equiv 2 \bmod 4$. Let $K = \mathbb{Q}(\xi_m)$ be a cyclotomic field and $L = K(\{\sqrt{u_{pq}}\}_{p<q\in S_m})$. Let X be the affine variety over $\mathbb{Q}$ defined by $N_{K/\mathbb{Q}}(\Xi) = \alpha \in \mathbb{Q}^\times$. Then the 2-torsion subgroup of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ is generated by all $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi, \chi)$, where χ runs through all characters in $\mathrm{Hom}(\mathrm{Gal}(L/K), \mathbb{Q}/\mathbb{Z}) \subset \mathrm{Hom}(\mathrm{Gal}(\bar{\mathbb{Q}}/K), \mathbb{Q}/\mathbb{Z})$.*

Proof. Let χ be a nontrivial character of $\mathrm{Gal}(\bar{\mathbb{Q}}/K)$. Then there is a subfield $L' \supset K$ of L with $[L' : K] = 2$ such that χ factors through $\mathrm{Gal}(L'/K)$. And $L'/\mathbb{Q}$ is Galois and non-abelian by [15, Theorem 1 and Proposition 1]. L'/K satisfies the condition $(*)$ follows from the fact:

suppose the group M is cyclic and there is the following central extension

$$
0 \to \mathbb{Z}/2\mathbb{Z} \to G \to M \to 0,
$$

then G is abelian.

Indeed, let $M = \langle\beta\rangle$ and $\mathbb{Z}/2\mathbb{Z} = \langle\gamma\rangle$. Let $\hat{\beta}$ be a fixed lifting of β in G, then we can see any element in G has the form $\hat{\beta}^i\gamma^j$ with $i, j \in \mathbb{Z}$. Note that γ is in the center

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of G, obviously G is abelian. Therefore, $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi,\chi) \in \mathrm{Br}(X^c)$ and is nontrivial by Theorem 5.

Let $d = \#S_m$. On the other hand, the Galois group $\mathrm{Gal}(L/K) \cong (\mathbb{Z}/2\mathbb{Z})^{d(d-1)/2}$ by the linear independent u_{pq} in $K^\times/K^{\times 2}$ (see [15, Lemma 4]). Therefore the subgroup of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ generated by all $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi,\chi)$ is of 2-rank $d(d-1)/2$. Using the Künneth formula [10, Chapter II, Section 1, Exercise 7], we can calculate that the 2-rank of $H^3(\mathrm{Gal}(K/k),\mathbb{Z})$ is also $d(d-1)/2$. Since $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c) \cong H^3(\mathrm{Gal}(K/k),\mathbb{Z})$, the 2-rank of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ is $d(d-1)/2$. Therefore all $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi,\chi)$ generate the 2-torsion subgroup of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$. □

Finally we will give an explicit example associated to a cyclotomic field.

Example 15. Let $K = \mathbb{Q}(\xi_{7\cdot 53})$. For any positive rational number α, we can write $\alpha = 2^{s_1}7^{s_2}53^{s_3}p_1^{e_1}\cdots p_g^{e_g}$, where $p_1,\dots,p_g$ are distinct primes which do not divide 742, and $s_1,\dots,s_3$ are integers, and $e_1,\dots,e_g$ are nonzero integers. Let $D(\alpha) = \{p_1,\dots,p_g\}$. Denote

$$D_1 = \left\{p \in D(\alpha):\ \left(\frac{-7}{p}\right) = \left(\frac{53}{p}\right) = -1\right\}$$

$$D_2 = \left\{p \in D(\alpha):\ \left(\frac{-7}{p}\right) = \left(\frac{53}{p}\right) = 1 \text{ and } \left(\frac{5+2\sqrt{-7}}{p}\right) = -1\right\}.$$

Then the equation $N_{K/\mathbb{Q}}(\Xi) = \alpha$ is solvable over $\mathbb{Q}$ if and only if the following conditions hold:

(i) The equation $N_{K/\mathbb{Q}}(\Xi) = \alpha$ is solvable over $\mathbb{Q}_p$ for each place p.
(ii) (BM-obstruction): $(-1)^{\sum_{p_i\in D_1} e_i/2+\sum_{p_i\in D_2} e_i} = (-1)^{s_2}\cdot(\frac{\alpha,-1}{2})$.

Proof. Let X be the affine variety defined by $N_{K/\mathbb{Q}}(\Xi) = \alpha \in \mathbb{Q}^\times$. We can see

$$\mathrm{Br}\big(X^c\big)/\mathrm{Br}_0\big(X^c\big) \cong H^3\big(\mathrm{Gal}(K/\mathbb{Q}),\mathbb{Z}\big) \cong \mathbb{Z}/2\mathbb{Z}.$$

It is easy to verify that $K(\sqrt{5+2\sqrt{-7}})/\mathbb{Q}$ is Galois. Let $L = K(\sqrt{5+2\sqrt{-7}})$. Let χ be the unique nontrivial character of $\mathrm{Gal}(\bar{\mathbb{Q}}/K)$ which factors through $\mathrm{Gal}(L/K)$. Then $\mathrm{Cor}_{K/\mathbb{Q}}(\Xi,\chi)$ is the unique generator of $\mathrm{Br}(X^c)/\mathrm{Br}_0(X^c)$ by Theorem 5. The proof follows from a similar argument as in Example 8. □

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